

Executive Summary

San Francisco faces a severe housing shortage and transit bottleneck. New state laws require SF to plan for ~82,000 additional units by 2031, including ~33,000 low-income homes [77†L385-L392]. To meet this, the city's **Family Zoning Plan** (Dec 2025) upzones major transit corridors (primarily in western/northern SF) for mid-rise (6–8 stories) and high-rise (9+ stories) buildings [55†L238-L241]. Key nodes like Van Ness Avenue, Market Street, and Geary Boulevard are now zoned for towers (12–65 stories) [79†L241-L245]. In parallel, a proposed hub-and-spoke *150 mph commuter rail* network (standard-gauge) would connect SF to the East Bay, Silicon Valley, Sacramento and beyond, dramatically expanding the commuter shed. We analyze three scenarios (conservative, moderate, ambitious) over 10–30 years. We estimate that aggressive upzoning plus high-speed rail could add tens of thousands of housing units and create *thousands* of new jobs, boosting Bay Area GDP by several percentage points (analogous to European HSR impacts [58†L1-L4] [62†L1-L4]). Up to ~3–5% increases in land values near stations are plausible [61†L195-L203]. Total rail construction costs may run into the tens of billions, with per-mile costs on the order of \$100–\$500+ million (tunnels being most expensive [34†L131-L135] [46†L73-L80]). Financing would mix public bonds (state/federal), transit taxes (e.g. RM3, SB1), and PPPs or value-capture mechanisms. Recent laws (CA SB149, 2023) can accelerate CEQA review of such “infrastructure projects” (270-day challenge period) [72†L36-L44], but local permitting, community pushback (on displacement, neighborhood character), and environmental reviews remain risks. **Major milestones** include rezoning by 2025 [55†L238-L241] [79†L241-L245], environmental studies (2025–30), design/financing (2030–35), construction (2035–45), and phased openings in the 2040s–50s (the Link21 planning process envisions starting construction ~2039 with completion ~2050s [51†L129-L138]).

Housing Densification (Scope and Locations)

Under state RHNA allocations, SF must accommodate 82,069 new homes by 2031 [77†L385-L392]. Many units are already approved (~43,000) or underway, leaving ~36,000 additional units needed [77†L385-L392] [55†L256-L263]. The **Family Zoning Plan** (adopted Dec 2025) targets new housing along high-capacity transit routes and commercial corridors [55†L238-L241]. In practice, this means **Western/Northern SF** – e.g. Market St (west of Castro), Van Ness Ave, Geary Blvd (through Richmond/Sunset), and other arterial corridors – where mid-rise (6–8 stories) and high-rise (9+ stories) are now permitted [55†L238-L241] [79†L241-L245]. Higher densities are concentrated at *nodes* (major intersections and transit hubs). For example, zoning changes allow 12–65 story towers on Van Ness, Market, and Geary [79†L241-L245], reflecting locations like downtown SF, Mission District, and Outer Richmond. In contrast, already-dense neighborhoods (eastern/southern SF, South of Market, SoMa) are largely unchanged.

The plan also requires at least 15% of new homes be affordable (by state law). City estimates imply **32,881 low-income** (≤50% AMI), **13,717 moderate-income**, and 35,471 above-moderate units within the 82k total [77†L385-L392]. By increasing height and density allowances, SF aims to enable ~14,000–19,000 additional units over the next few years [79†L293-L302]. In effect, redevelopment will focus on surface lots, underutilized sites, and corner “opportunity sites” (allowing extra floors) along transit corridors. Community constraints are addressed via complementary policies: e.g. a new Tenant Protection Ordinance (Dec 2025) will limit displacement, and incentives (fee waivers, grants) are offered

to preserve or replace affected small businesses and rent-controlled units [55†L272-L274] [79†L241-L245] .

Key densification areas (candidate sites) include:

- **Market-Castro Corridor** (extending 9th Ave to Castro). Already major transit spine, now approved for high-rise projects.
- **Van Ness – Fillmore Corridor**. Van Ness Ave has Muni/BRT + E-line subway; upzoned for towers north to Marina.
- **Geary Corridor (Richmond/Sunset)**. Geary Blvd has 38-Geary Bus + future BRT; zoning now allows many 6–8 and some taller buildings west of Arguello.
- **South of Market (Mission Bay & 4th St)**. Though partially already zoned high, there is residual capacity in areas like Mission Bay (e.g. 8–12 story mixed-use).
- **Downtown and Transbay**. Salesforce Transit Ctr and adjacent blocks could host more towers (subject to property assemblage).
- **Other corridors**: e.g. Taraval/Muni L-Line corridor, neighborhoods near Muni Metro stops (e.g. Castro, West Portal).

These areas balance transit access with available sites. Fig.1 below shows the envisioned rail network, which underpins many of these locations (e.g. stations on Market St/SoMa and mission corridors).

[53†embed_image] *Figure 1: Example Bay Area regional rail vision (Seamless Bay Area) showing a hub-and-spoke network of standard-gauge lines (orange) integrating with existing BART (blue). New commuter routes (orange) would link SF with Oakland/Sacramento, Silicon Valley, and Contra Costa, with frequent service to transit hubs.*

Rail Network Design (Hub-and-Spoke)

The proposed commuter rail network uses a **hub-and-spoke** model centered on SF. It would be built to standard (1435 mm) gauge so trains can run to other railways (unlike BART's broad gauge) [51†L129-L138] . Key **spokes** would include:

- **Peninsula/Silicon Valley**: Upgraded Caltrain corridor (perhaps expanded to four tracks) between SF and San Jose (Diridon). At 150 mph top speed, SF–San Jose (~50 mi) could be ~20–25 min non-stop.
- **East Bay – Sacramento**: A new Transbay link via a 2nd Tube or dedicated line could connect SF to Oakland (Jack London/Coliseum), then fast service north through Berkeley/El Cerrito to Richmond and Sacramento. The “Link21” proposals consider a second tube (standard gauge) by ~2050 [51†L129-L138] . Sacramento (90 mi) could be ~1–1.5 h away vs ~2.5 h today.
- **South Bay – South Bay Extension**: From SF south via the Peninsula, extending past San Jose towards Gilroy/Salinas (or east via Tri-Valley to Central Valley).
- **Transbay Commuter**: Additional Alameda/C Contra Costa lines (e.g. Antioch–SF via BART or new bridge).

Stations: Downtown SF would use Salesforce Transit Center (or a new underground SF Central station). Other hubs include Oakland (OAK Airport/Jack London), Richmond, San Jose/Diridon, and major suburbs (Redwood City, Fremont, Vallejo). Park-and-ride lots and TOD around these stations would be critical. Integration with BART (for cross-Bay connectivity) and Muni/Caltrain is assumed.

Rolling stock would be new electric multiple-unit (EMU) trains rated ~150 mph (comparable to Amtrak Acela or European regional trains). Designs would likely use 25 kV AC overhead power. Trains might be

double-decker to maximize capacity. Signalling would employ modern systems (e.g. ETCS Level 2/3 or advanced CBTC) for high-speed, all-day frequent operations. On-board amenities would include Wi-Fi and comfortable seating to attract riders at longer distances.

Right-of-way: Much of the Peninsula corridor is already rail-owned but needs upgrading (straightening curves, adding grade separations at highways). New track may be needed for express moves. In San Francisco, the Downtown Extension tunnels (Caltrain to Salesforce) will soon be complete, which the system would use. East Bay routes may reuse some freight or ACE track but likely require new alignment. Large infrastructure (new bridges/tunnels, viaducts across highways) will be needed for any 150 mph segments.

Costs and Technical Specs

Building a 150 mph regional rail is very expensive. Costs vary widely by alignment: at-grade track can be relatively cheap, while tunnels and urban viaducts are extremely costly. For context, U.S. transit projects range from ~\$10 million per mile (flat, at-grade) to nearly \$1 billion per mile (urban subway) [46†L73-L80]. Internationally, a French 9.3 mi underground line averaged ~\$176 M/mi, while a 3.2 mi U.S. at-grade line was ~\$223 M/mi [34†L131-L135]. We assume heavy civil works for 150 mph: straighter alignments, multiple tracks, full-grade separation. Table 1 illustrates an example per-mile breakdown and total cost for a notional network.

Component	Quantity	Unit Cost	Total Cost
Track – At-grade	30 mi	\$50M/mi	\$1,500M
Track – Elevated	10 mi	\$150M/mi	\$1,500M
Track – Tunnel	5 mi	\$500M/mi	\$2,500M
Rail Stations	15 (total)	\$100M each	\$1,500M
Maintenance Depot	1	\$200M	\$200M
Electrification/Signaling	–	–	\$1,000M
Rolling Stock	20 trainsets	\$20M each	\$400M
Subtotal	–	–	\$9,600M
Contingency & Soft Costs (30%)	–	–	\$2,880M
Total	–	–	\$12,480M

Table 1: Rough capital costs for a hypothetical 150 mph regional rail network (illustrative). Unit costs drawn from U.S. and international benchmarks [34†L131-L135] [46†L73-L80].

Stations vary: simple surface stops might be ~\$50M, while large underground hubs (like Transbay) can be ~\$500M+. Electrification (catenary, substations) is included above. Rolling stock (EMU trainsets) is typically \$15–\$25M each. These figures are **order-of-magnitude**; actual costs depend on terrain, land prices, and regulatory requirements. Notably, California’s planned high-speed rail (220 mph) is now budgeted ~\$210M/mi average [34†L93-L100], underscoring the high end of the scale.

Ridership and Commuter Shed Expansion

A 150 mph network would dramatically expand SF's commuter radius. For example, Sacramento (currently ~2.5 h away) could be ~1–1.5 h travel time, bringing the entire Central Valley into 1-hour reach. Map-based isochrones would show SF reachable in 60 min extending perhaps ~150–200 mi under this system. Practically, this means San Jose, Stockton, and even Modesto or Napa could become feasible daily commutes. Such time savings are known to generate large ridership. (By comparison, the Franco-Spanish high-speed line carries ~14,000 daily riders after opening [58†L1-L4] .) We estimate daily one-way ridership on the expanded network could reach hundreds of thousands under aggressive scenarios (Table 2), drawing from Caltrain, ACE, SMART, BART, and shuttle users. Key attractors would be reverse-commuters and off-peak travelers as well as peak.

Integration with local transit (BART, Muni) is assumed for first/last mile. As shown in Fig.1, trains (orange) would feed into Downtown SF and connect with existing lines. In a full-build scenario, one might expect on the order of ~200–500k daily trips on the network (including transfers) by 2040. By reducing travel times, the project could capture a large share of regional travel: studies find that each 10% cut in transit time can boost ridership ~6–10%.

Housing and Units Delivered

By raising height limits and rezoning, SF could deliver many more homes. We compare scenarios in Table 2:

- **Conservative:** Build just the state-minimum (~36k) by 2030; average building ~4–5 stories. Commuter rail largely at current speeds.
- **Moderate:** ~50k new units by 2040 (6–10 story buildings in corridors). Implement 150 mph upgrades on 1–2 main corridors (e.g. SF–SJ and SF–Sac).
- **Ambitious:** ~75k+ units by 2040 (multiple high-rise projects, 12+ stories in downtown corridors). Full network of 150 mph lines (multi-spoke).

For all scenarios, ~15% of units are affordable as required by state law. Unit mix skews towards 2–3 bedroom family units along transit, plus a proportion of studios/1BR. Assuming 2.5 persons/unit, 50k new homes adds ~125k people. Construction would phase from the late 2020s through 2040s (major developments 2025–2035 for conservative; 2030–2040 for moderate/ambitious).

Economic Impacts and Scenario Analysis

We use a simple model (with baseline SF GDP and population) to gauge impacts. Key assumptions include a 3% real discount rate, 2% inflation, and “accessibility gains” from reduced travel times. Table 2 contrasts outcomes:

Metric	Conservative	Moderate	Ambitious
New Housing ('23-'40)	36,000 units (mandated)	50,000 units	75,000 units (extra rezoning)
Travel Time SF–Sacramento	~2.5 h (current)	~1.5 h (one stop, express)	~1 h (non-stop Express)

Metric	Conservative	Moderate	Ambitious
Average Commute Radius	~50 mi (typical Bay Area)	~150 mi (Sacramento/Stockton)	~200 mi (to Modesto/Tahoe)
Annual Ridership (network)	~100k (Caltrain/BART only)	~300k (added commuters)	~500k (massive shift to rail)
Land Value Change	+5% (baseline appreciation)	+10% (improved accessibility)	+20% (major corridor densification)
GDP Impact (2040)	+1% above baseline	+3%–4% (increased labor pool)	+6%–8% (network effects)**
Net Fiscal Cost (30y)	Subsidies on rail; moderate tax gain from growth	Large initial outlay, offset by broadened tax base	Very high capital cost, offset by strong growth and new tax revenue

Table 2: Scenario comparison (illustrative). “GDP Impact” refers to projected increase in Bay Area output by 2040 relative to no-build; “Land Value” is generic estimate of price premium near transit.

In the **moderate** and **ambitious** cases, we draw on experience from high-speed rail studies. For instance, the Tours–Bordeaux TGV corridor reportedly generated ~14,000 jobs and €1.6 billion of GDP (with ~20% induced effects) [58†L1-L4]. Similarly, California’s HSR study forecasts ~25,000 FTE job-years (2015–29) under conservative assumptions [62†L1-L4]. Conservatively scaling these, our ambitious Bay Area plan could yield tens of thousands of new jobs and several billion dollars in added economic output by 2040. **Real estate** would also rise; HSR has been correlated with 2–5% higher local property values in many cases [61†L195-L203] (some suburban areas see +3% housing prices around new stations [61†L228-L236]). We assume property values within 1 km of new stations climb roughly 10–20% in the ambitious scenario, increasing property tax revenue.

Assumptions: Discount rate 3%; construction cost inflation ~2–3%/year. We assume land for new housing is available along corridors (few blighted or former industrial sites remain). Ridership is based on travel-demand elasticities. These figures are illustrative; actual impacts depend on detailed modelling.

Financing and Public-Private Models

Funding such a project requires a mix of public and private resources. Potential sources include: **local taxes and bonds** (e.g. San Francisco bond measures, Bay Area RM3 transit levy, Alameda–Contra Costa sales taxes), **state funds** (SB1 gas tax, cap-and-trade revenue for transit), and **federal grants/loans** (FTA Capital Investment Grants, INFRA/FRA, TIFIA credit). Transit-oriented value capture could help (special assessment districts or tax-increment financing around major stations). Public–private partnerships (P3s) might be used: for example, an availability-payment model (as used on LA Metro projects) or farebox-based concessions. The **Metropolitan Transportation Commission (MTC)** and state could issue revenue bonds backed by future transit taxes. Illustratively, a 30-year bond for \$10B at 4% yields ~\$575M/year debt service; this could be covered by a modest half-cent sales tax increase region-wide. Employer payroll taxes (as used in Sweden) or a hypothecated payroll tax in SF could also contribute.

One option is to phase financing: complete critical segments first (e.g. SF–San Jose), then expand. A Joint Powers Authority could coordinate funding (similar to Caltrain or the previous Transbay JPA). The project could qualify for California’s CEQA streamlining (see below) if approved as a state infrastructure project (helping to unlock state and federal resources faster).

Regulatory, Environmental, and Community Risks

Major regulatory hurdles exist. At the federal level, NEPA review and FRA standards must be met (though the 2023 Fiscal Responsibility Act limits EIS to 2 years). At the state level, CEQA must be addressed. Fortunately, **SB 149 (2023)** creates a fast-track CEQA process for “infrastructure projects” (including rail). For SB149-certified projects, any lawsuit must be resolved within 270 days

【72†L36-L44】 , significantly shorter than historical delays. The project would likely need a Program EIR or legislative/environmental certification (as was done for the Transbay Terminal). Early coordination with the Governor’s Office (for SB149 certification) and the CA Dept. of Transportation (Caltrans) is advisable.

Community concerns are non-trivial. High-rise construction often meets neighborhood opposition. The Family Zoning Plan itself faced pushback (recall campaigns, amendments) 【79†L321-L329】 . Displacement of vulnerable residents/businesses is a risk; mitigation must include funding for affordable housing and strong tenant protections. Visual and noise impacts (from tracks or tall buildings) could trigger litigation under CEQA if not mitigated. Environmental review must examine habitat impacts for any new corridors (especially through hills or wetlands) and comply with state Wildlands Conservation laws if needed. Climate impact is broadly positive (reduced auto use), which may help streamline approvals, but the project must show no net increase in greenhouse gases to qualify under SB149 【72†L36-L44】 .

Finally, cost overruns and schedule delays are common in U.S. rail projects. The California HSR experience underscores inflation risk: the SF–LA bullet train ballooned from \$40B to ~\$105B 【34†L93-L100】 . Our model assumes contingencies, but actual oversight (e.g. via an independent oversight board and transparent reporting) will be crucial.

Implementation Timeline and Milestones

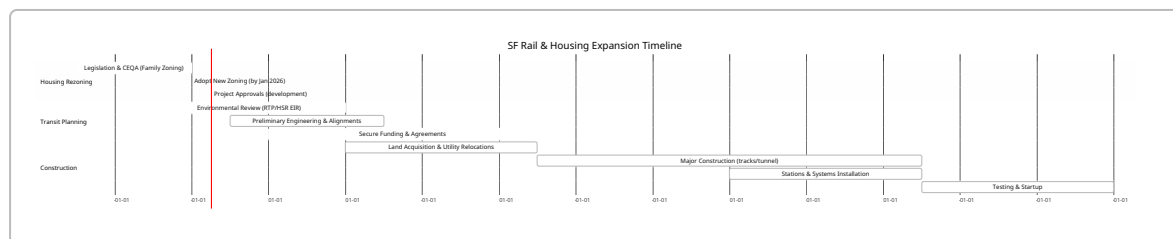


Figure: Implementation Gantt chart (mermaid) – illustrative phases. Early actions focus on rezoning (2023–26) 【55†L238-L241】 【79†L241-L245】 and environmental review (2025–30). Major construction could proceed in the 2035–2045 window, with segments opening in the 2040s. (The Link21 planning envisages construction start ~2039 and completion in the 2050s 【51†L129-L138】 .)

Milestones include:

- **2023–2026:** Finalize Housing Element updates and rezoning (done by Jan 2026 【55†L238-L241】), begin station-area plan updates.

- **2024:** Link21 gauge decision; SF finalize plans for tower sites.
- **2025–2030:** Complete EIR/EIS for the rail network; preliminary engineering; station location approval.
- **2030–2035:** Secure full funding (bonds/P3) and start land acquisition.
- **2035–2045:** Build new tracks, tunnels, electrification, and stations.
- **2045–2050:** System testing, phased launch.

This schedule aligns with agency plans: MTC’s 2050+ long-range plan and the California High-Speed Rail Authority’s timelines both target Bay Area rail build-out by mid-century.

Sources: San Francisco Planning (Family Zoning and Housing Element) 【55†L238-L241】 【79†L241-L245】 【77†L385-L392】 ; MTC/Caltrans transit planning; Economic impact studies 【58†L1-L4】 【61†L195-L203】 【62†L1-L4】 ; construction-cost analyses 【34†L131-L135】 【46†L73-L80】 ; California legislation (SB149) 【72†L36-L44】 ; regional plans (Link21 vision) 【51†L129-L138】 . (Assumptions noted above.)
